
Summary

Write your summary here...

Sammendrag

Skriv sammendraget ditt her...

Preface

Write your preface here...

Table of Contents

Summary	i
Sammendrag	ii
Preface	iii
Table of Contents	vi
List of Tables	vii
List of Figures	ix
Abbreviations	x
1 Introduction	1
1.1 Motivation	1
1.2 Previous Work	2
1.3 Problem Description	2
1.4 Use of AI Tools	2
1.5 Contributions	3
1.6 Outline	3
2 Literature Review	5
2.1 Historical Background	5
2.2 Model Predictive Control for Vehicle Path Following	6
2.2.1 Vehicle Models Used in MPC-Based Path Following	7
2.2.2 MPC Formulation for Path Following	7
3 Background Theory	9
3.1 Vehicle Dynamics	9
3.1.1 Kinematic Model	9
3.1.2 Dynamic Model	10

3.1.3	Tire Model	10
4	Method	13
5	Results	15
6	Conclusion	17
	Bibliography	19
	Appendix	21

List of Tables

List of Figures

3.1 (placeholder for) Kinematic bicycle model	10
---	----

Abbreviations

Symbol = definition

Chapter 1

Introduction

1.1 Motivation

Hva er motivasjonen for å jobbe med selvkjørende biler? Hvorfor er det viktig?

- Samfunnsmessig motivasjon: menneskelige feil står for majoriteten av trafikku-lykker. Automatiserte systemer kan redusere ulykker
- Effektivitet og mobilitet: bedre flyt, mindre kø, mer forutsigbar kjøring
- Tilgjengelighet: økt mobilitet for eldre og personer uten førerkort

Nevn FNs bærekraftsmål: mål 9 og 11

- Goal 9: Industry, Innovation and Infrastructure
- Goal 11: Sustainable Cities and Communities

Evt. nevnt relevans for kontrollsystemer i selvkjørende biler og knyttet det til oppgaven

- Societal:

Transportation systems are undergoing a profound transformation driven by advances in automation, artificial intelligence, and control engineering. Autonomous vehicles (AVs) are widely regarded as one of the most transformative technologies within modern mobility, with the potential to significantly improve traffic safety, efficiency, accessibility, and environmental sustainability.

From a societal perspective, road traffic accidents remain a major global challenge. Human error is a contributing factor in the vast majority of accidents, often caused by distractions, fatigue, misjudgment, or impaired driving. By transferring critical driving functions from human to automated systems, autonomous vehicles have the potential to substantially reduce accident rates, and improve road safety. In addition, automated driving enables smoother velocity profiles, reduced stop-and-go traffic,

and optimized energy consumption, which may contribute to lower emissions and more sustainable transportation.

Autonomous mobility also plays an important role in improving accessibility. Reliable self-driving systems can provide transportation services to elderly individuals, people with disabilities, and others who are unable to drive themselves. In this way, autonomous vehicles are not only a technological innovation, but also a tool for increasing social inclusion and mobility equality.

- **Technological:**

Despite these promising benefits, achieving reliability and scalable autonomy remains a significant engineering challenge. Autonomous driving requires the integration of perception, localization, planning, and control into a cohesive real-time system operating under uncertainty. Among these components, vehicle control plays a fundamental role. Even with accurate perception and planning, the vehicle must be capable of executing precise and stable motion in order to safely follow a desired path.

- This master's thesis is connected to the United Nations Sustainable Development Goals (**SDGs**), particularly Goal 9: Industry, Innovation and Infrastructure and Goal 11: Sustainable Cities and Communities. Goal 9 highlights the need for resilient infrastructure and continuous technological innovation. The development and evaluation of advanced control strategies for autonomous vehicles contribute to progress within intelligent transportation systems.
- The project is also relevant to Goal 11, which promotes safer, more sustainable, and inclusive urban environments. Since human error is responsible for the majority of traffic accidents, improvements in automated driving technologies may play an important role in enhancing road safety. More accurate and efficient control systems can also support smoother traffic flow and improved energy utilization.

1.2 Previous Work

Hva er det som har blitt gjort tidligere. Generelt eller på NapLab-bilen.

1.3 Problem Description

Hva er problemstillingen masteroppgaven tar for seg og vil løse.

1.4 Use of AI Tools

Hvordan har AI blitt brukt i oppgaven. Hvilke midler og hvordan.

- chatGPT - kode, skriving
- copilot - rettskriving + finskriving

1.5 Contributions

Hva er det masteren bidrar med. Hva er gjort og hva er nytt.

1.6 Outline

Oversikt over hvert kapittel og hva de inneholder.

Chapter 1 Introduction

Chapter 2 Literature Review

Chapter 3 Background Theory

Chapter 4 Methods

Chapter 5 Results

Chapter 6 Discussion

Literature Review

2.1 Historical Background

Hva er blitt gjort på selvkjørende biler hittil. Kort historisk gjennomgang. DARPA, Waymo etc. (Peder hadde en kilde på dette i hans master)

- Foundational research
- Grand Challenges
- Commercial Development (Waymo)
- State today

Early research on autonomous vehicles between 1980 and 2000 laid the technological foundation for later breakthroughs. During this period, university research centers, often in collaboration with automotive manufacturers and transportation agencies, explored two primary development paths. One line of research focused on automated highway systems, where vehicles relied on dedicated infrastructure for guidance. A notable demonstration took place in 1997 on California's I-15 highway, where the PATH program successfully demonstrated vehicle platooning using magnets embedded in the roadway combined with vehicle-to-vehicle communication (Samaras et al.).

A second line of research pursued autonomous vehicles that relied primarily on on-board perception and control systems rather than infrastructure support. In the early 1980s, a research team led by Ernst Dickmanns at the Bundeswehr University Munich developed a vision-guided vehicle capable of navigating highways at speeds of up to 100 km/h under traffic-free conditions. Similar efforts were carried out at Carnegie Mellon University through the NavLab projects, which demonstrated long-distance autonomous steering in the 1990s. Although these projects demonstrated significant technical progress, limitations in sensing technology, computational power, and system robustness prevented large-scale deployment at the time (Samaras et al.).

A major turning point occurred with the DARPA Grand Challenges competitions. The 2004 challenge, focused on off-road autonomous navigation, resulted in no vehicles completing the course. However, by 2005, several teams successfully finished the desert route, demonstrating rapid technological progress (Buehler et al., 2009). The 2007 DARPA Urban Challenge further advanced the field by requiring vehicles to operate in urban environments with traffic rules and dynamic obstacles. The competition showcased the feasibility of combining perception, localization, and decision-making in complex settings, and many participants later became leaders in commercial autonomous driving development.

In 2009, Google launched its self-driving car project, which later became Waymo. This marked the beginning of large-scale industrial investment in autonomous vehicles (Waymo). Around the same time, major automotive manufacturers and technology companies began investing heavily in advanced driver assistance systems (ADAS), such as adaptive cruise control and lane-keeping assist. The Society of Automotive Engineers (SAE) introduced a standardized taxonomy for driving automation levels (SAE J3016), ranging from level 0 (no automation) to level 5 (full automation). (International, 2018). Most production vehicles today operate at level 1 or level 2, requiring continuous human supervision. By the mid 2010s, autonomous taxi pilot programs emerged. Waymo launched limited commercial robotaxi services in geofenced urban areas in the United States, demonstrating driverless operation under controlled operational design domains (ODDs)(?) waymo.

In recent years, progress has focused on improving robustness, redundancy, and scalability. Companies such as Waymo, Cruise, Baidu, and others have deployed commercial robotaxi services in selected cities. Meanwhile, Tesla has developed advanced driver assistance features under its "Full Self-Driving" (FSD) branding, although these systems remain classified as level 2 automation under SAE standards. Despite technological advances, fully autonomous driving in unrestricted environments remains an open research challenge. Today, research continues to focus on improving perception systems, sensor fusion, vehicle modeling, motion planning, and control.

2.2 Model Predictive Control for Vehicle Path Following

Model Predictive control (MPC) is a control strategy in which an optimal control problem is solved repeatedly over a finite prediction horizon using a model of the system dynamics. Due to this optimization-based structure, MPC is particularly well suited for control problems where future reference information is available and physical constraints must be explicitly respected. In the context of autonomous vehicles, path following is naturally suited to this framework, as the vehicle is required to follow a predefined reference path while satisfying constraints on steering, steering rate, velocity and vehicle dynamics.

As a result of these properties, MPC has become one of the most widely used control frameworks for path following in autonomous vehicles. Numerous studies have demonstrated its effectiveness in both simulation and experimental settings, making MPC a natural choice for mid-level control tasks in autonomous driving systems.

2.2.1 Vehicle Models Used in MPC-Based Path Following

The choice of vehicle models plays a central role in the design of MPC schemes for path following, as it directly affects prediction accuracy, computational complexity, and real-time feasibility. In existing literature, a wide range of vehicle models has been employed, ranging from simple kinematic representations to more advanced dynamic models incorporating tire forces and actuator dynamics.

Kinematic models are frequently adopted in MPC-based path following applications due to their favorable trade-off between model simplicity and computational efficiency. Tang et al. (2020) present an improved kinematic MPC for high-speed path tracking based on a kinematic bicycle model, where yaw rate is used as the control variable instead of the steering angle. To reduce model mismatch caused by tire slip at higher speeds, an analytical sideslip angle compensation derived from steady-state vehicle dynamics is introduced. The resulting controller is implemented in a cascaded structure, with an outer MPC generating a desired yaw rate and an inner **PID** controller tracks this reference via steering actuation. Despite the improved tracking performance, the approach remains fundamentally kinematic, as no lateral vehicle dynamics or tire force prediction is included in the MPC.

To improve prediction accuracy in scenarios where lateral dynamics become significant, several studies employ dynamic vehicle models within the MPC framework. Kim et al. (2014) propose an MPC-based path-following controller using a linearized dynamic bicycle model augmented with steering actuator dynamics. By including lateral velocity and yaw-rate dynamics, the controller is able to represent transient vehicle behavior more accurately than kinematic approaches. At the same time, the linearized formulation allows for more efficient computations. Similarly, Domina and Tihanyi (2023) use a linearized dynamic single-track model formulated in path-relative error coordinates. Vehicle speed is treated as a scheduling parameter, resulting in a **LPV** MPC formulation, thereby enabling the controller to account for speed-dependent dynamics.

At a higher level of model fidelity, nonlinear dynamic models have been adopted to enable autonomous driving close to the limits of handling. Liniger et al. (2015) employ a nonlinear dynamic bicycle model with nonlinear tire force characteristics, enabling accurate prediction of vehicle behavior under aggressive maneuvers. The nonlinear dynamics are locally linearized along the predicted trajectory at each time step, resulting in a linear time-varying MPC formulation. Although this approach provides high prediction accuracy and supports performance near handling limits, it also incurs increased computational complexity compared to linear and kinematic formulations.

2.2.2 MPC Formulation for Path Following

Despite differences in model fidelity and application domain, the cost functions employed in MPC-based path control exhibit several common characteristics across the reviewed studies. Tang et al. (2020), Kim et al. (2014) and Domina and Tihanyi (2023) all formulate quadratic tracking-based cost functions that penalize lateral position error and heading error with respect to a geometric reference path. This consistent use of geometric error metrics highlights the central role of lateral path accuracy in MPC-based vehicle control. Although Liniger et al. (2015) adopt a contouring-based formulation, the contouring and

lag errors used in their objective serve a similar purpose by penalizing geometric deviation from the reference path.

In addition to tracking accuracy, all reviewed approaches explicitly penalize control effort or variations in the control input. Tang et al. (2020) include penalties on yaw rate and yaw-rate increments, Kim et al. (2014) penalize incremental steering commands while Domina and Tihanyi (2023) apply penalties on steering angle magnitude. Liniger et al. (2015) also include changes in control inputs to ensure smooth and physically feasible vehicle behavior.

Across the reviewed studies, the formulation of MPC for path following differs primarily in the choice of system states and control inputs. Tang et al. (2020) formulate the MPC using the vehicle heading and lateral position as states, while path-following errors are introduced implicitly through subtraction of the reference trajectory. In contrast, Kim et al. (2014) and Domina and Tihanyi (2023) employ explicit path-relative error representation. Liniger et al. (2015) adopt a full vehicle-state representation based on the nonlinear dynamic bicycle model, where global position, heading angle, velocities, and yaw rate are included as states. In the contouring-based MPC formulation, the state vector is augmented with a path progress variable, enabling explicit optimization of advancement along the reference path.

Regarding control inputs, Tang et al. (2020) use yaw rate as the MPC input within a cascaded control structure, while Kim et al. (2014) optimize incremental steering commands, Domina and Tihanyi (2023) directly manipulate the steering angle, and Liniger et al. (2015) employ steering and longitudinal actuation, with an additional virtual input governing path progress.

MPC formulations differ in how constraints are incorporated, reflecting trade-offs between computational efficiency, physical feasibility, and safety. Tang et al. (2020) apply a simple constraint structure, imposing bounds on the yaw rate and its rate of change, while no explicit constraints are applied to the system states. Kim et al. (2014) introduce explicit constraints on steering angle and steering rate, both defined as functions of vehicle speed, to account for actuator limitations and safety-related lateral acceleration limits. Domina and Tihanyi (2023) similarly rely on input constraints by bounding the steering angle, and extend the formulation with steering rate constraints when actuator dynamics are included.

In contrast, Liniger et al. (2015) employ a richer constraint formulation tailored to high-performance driving. In addition to input bounds, explicit constraints are imposed on vehicle position to ensure that the vehicle remains within track boundaries. These geometric constraints are formulated as soft constraints, and additional bounds are applied to selected vehicle states to maintain feasibility during aggressive maneuvers.

- Types of MPC

Background Theory

3.1 Vehicle Dynamics

Vehicle dynamics provides the mathematical description of a vehicle's motion in response to steering, acceleration, and braking inputs. These models form the foundation for analysis, simulation, and control design in autonomous driving applications. Depending on the intended use, vehicle models may range from simple kinematic descriptions to more advanced dynamic formulations that account for forces, inertia, and tire behavior.

3.1.1 Kinematic Model

Kinematic vehicle models describe the motion of a vehicle based purely on geometric relationships between position, orientation, velocity, and steering input. These models neglect dynamic effects such as tire slip, inertia, and external forces, and instead assume pure rolling motion of the wheels. As a result, kinematic models are most accurate at low to moderate speeds and under smooth driving conditions.

A widely used kinematic representation is the kinematic bicycle model, in which the four wheels of the vehicle are reduced to a single front wheel and a single rear wheel located along the vehicle centerline. The model state typically consists of the vehicle position (x, y) , heading angle ψ , and longitudinal velocity v , while the steering angle δ serves as the control input. The vehicle geometry is characterized by the wheelbase L . The continuous-time kinematic bicycle model is given by:

$$\dot{x} = v \cos(\psi) \quad (3.1)$$

$$\dot{y} = v \sin(\psi) \quad (3.2)$$

$$\dot{\psi} = \frac{v}{L} \tan(\delta) \quad (3.3)$$

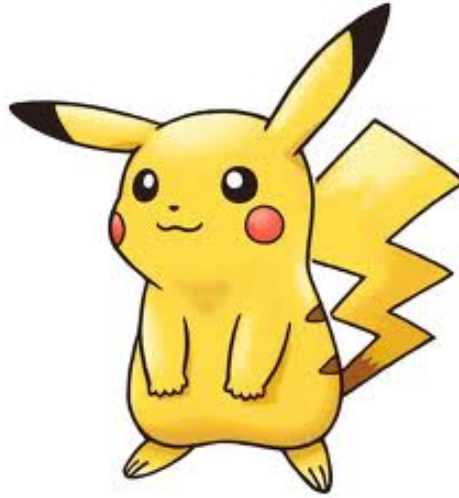


Figure 3.1: (placeholder for) Kinematic bicycle model

3.1.2 Dynamic Model

Dynamic vehicle models extend kinematic formulations by explicitly accounting for the forces acting on the vehicle and the resulting accelerations. In addition to geometric relationships, these models include the effects of vehicle mass, inertia, and tire–road interaction, enabling a more accurate description of vehicle motion over a wider range of operating conditions.

3.1.3 Tire Model

Tire models play a crucial role in vehicle dynamics, as tire–road interaction governs the generation of longitudinal and lateral forces required for motion and control. In lateral vehicle control and path-following applications, tire models are primarily used to describe how lateral forces depend on slip angle, vertical load, and road friction conditions.

Linear Model

For control-oriented modeling, simplified tire representations are commonly employed to balance physical realism and computational efficiency. A widely used approach is the linear tire model, where the lateral tire force is assumed proportional to the slip angle through the cornering stiffness. This approximation is valid for small slip angles and moderate lateral accelerations. Despite its simplicity, the linear tire model captures the dominant behavior of lateral force generation around nominal operating points and is well suited for real-time control design.

Dugoff's Tire Model

Pacejka Tire Model

For applications involving larger slip angles or operation closer to the limits of handling, nonlinear tire models are required. Empirical models such as the Pacejka Magic Formula describe tire forces using nonlinear functions fitted to experimental data, enabling accurate prediction of saturation effects and force coupling. While these models provide high fidelity, they significantly increase model complexity and computational load.

Chapter 4

Method

Chapter 5

Results

Chapter 6

Conclusion

Bibliography

- Buehler, M., Iagnemma, K., Singh, S., Siciliano, B., Khatib, O., Groen, F. (Eds.), 2009. The DARPA Urban Challenge: Autonomous Vehicles in City Traffic. volume 56 of *Springer Tracts in Advanced Robotics*. Springer Berlin Heidelberg, Berlin, Heidelberg. doi:10.1007/978-3-642-03991-1.
- Domina, Á., Tihanyi, V., 2023. Model Predictive Controller Approach for Automated Vehicle's Path Tracking. *Sensors* 23, 6862. doi:10.3390/s23156862.
- International, S., 2018. Taxonomy and Definitions for Terms Related to Driving Automation Systems for On-Road Motor Vehicles. Technical Report SAE J3016. SAE International. Warrendale, PA.
- Kim, E., Kim, J., Sunwoo, M., 2014. Model predictive control strategy for smooth path tracking of autonomous vehicles with steering actuator dynamics. *International Journal of Automotive Technology* 15, 1155–1164. doi:10.1007/s12239-014-0120-9.
- Liniger, A., Domahidi, A., Morari, M., 2015. Optimization-Based Autonomous Racing of 1:43 Scale RC Cars. *Optimal Control Applications and Methods* 36, 628–647. doi:10.1002/oca.2123, arXiv:1711.07300.
- Samaras, C., Oluwatola, O.A., Corporation, RAND., . Brief History and Current State of Autonomous Vehicles .
- Tang, L., Yan, F., Zou, B., Wang, K., Lv, C., 2020. An Improved Kinematic Model Predictive Control for High-Speed Path Tracking of Autonomous Vehicles. *IEEE Access* doi:10.1109/ACCESS.2020.2980188.

Appendix

Write your appendix here...